Curriculum Vitae

Personal Information

Name	**Juan Maria Carvajal Ramos**
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Professional Summary

Senior Naval Architect with 15+ years of experience in **marine engineering, energy systems,** and **decarbonisation technologies**. Expertise in **hydrodynamic analysis, CFD modelling, and propulsion systems design**, including **alternative fuels, hydrogen-based and waste-heat recovery solutions**. Extensive experience in applying **international standards (SOLAS, MARPOL, DNV)** and conducting **technical assessments of complex engineering systems**. Proven ability to **analyse, interpret, and document technical information** with clarity and precision, including the evaluation of novel engineering concepts and technologies.

Professional Experience

Orcan Energy AG – Munich, Germany

Project Manager & Product and Application Engineer | Sep 2023 – Aug 2025

- Developed **ORC-based Marine Efficiency PACKs**, optimising vessel performance and CO_2 emissions reduction, improving **EEDI, EEXI, and CII ratings** with typical 2–4 year ROI.
- Advised on **integration of ORC systems** in marine environments, assessing technical feasibility and performance.
- Analysed vessel operational data in secure environments to generate actionable performance insights.
- Oversaw **project simulations, heat transfer assessments, and component selection.**
- Coordinated c**ertification processes** with DNV, BV, ABS, LR, and RINA.
- Led **cross-functional engineering teams** (mechanical design, automation, thermodynamics) to deliver **integrated marine energy systems.**

Key projects: Marine Efficiency PACKs – eP M 050.100 / eP M 150.2

Hydrogenious LOHC Technologies GmbH – Erlangen, Germany

Senior Naval Architect & Mechanical Design Engineer | Apr 2020 – Jun 2023

- Designed **hydrogen-based propulsion systems** integrating LOHC technology, fuel cells (PEMFC/SOFC), and ship power systems.
- Evaluated **compliance and safety** of hydrogen-based propulsion systems against maritime regulations (IGF, SOLAS, MARPOL, DNV), including HAZID and fire safety scenarios.
- Led **technical documentation and approval processes,** resulting in **Approval in Principle** (AiP) and **Certification of Materials and Components** for hydrogen-based marine systems.
- Defined and monitored **technical KPIs** (e.g., footprint, mass, energy density, conversion efficiency) to drive optimisation and track system maturity.
- Coordinated **prototyping, testing, and supplier evaluation for PEM fuel-cell systems**, ensuring regulatory compliance and performance standards.

Key Projects: 200kW LOHC and fuel cell power test unit for Offshore Supply Vessel (HyNjord-project), and a megawatt-scale commercial product (EU-funded Ship-Ah2oy project)

BlueNewables – Santa Cruz de Tenerife, Spain

Professional consultant / Senior Naval Architect *(Part-time, Remote)* | Oct 2019 – Jun 2021

- Provided **technical consultancy** on offshore hydrogen production and **renewable energy integration** (wind + H_2).

Navitecture GmbH – Hamburg, Germany

Senior Naval Architect / Marine Engineer *(Part-time, Remote)* | Oct 2018 – Mar 2020

- Led **ship design initiatives** including **hull optimisation using CFD** (2-5% efficiency improvements) and **hybrid propulsion analysis,** including hydrogen fuel cell applications.

Key Projects: FS35-Uthörn (Methanol-powered research vessel), MPV70 Ecuador

Parental leave | Apr 2018 – Sep 2018

Sea Master Consulting & Engineering – El Puerto de Santa Maria, Spain

Project Manager / Senior Naval Architect / R&D Engineer | Nov 2016 – Mar 2018

- Managed **offshore platform mooring systems** including performing hydrodynamic assessments using CFD and MetOcean data, optimising mooring configurations, and ensuring offshore platform integrity in harsh environments.
- Managed **technical ROPAX ferry design**, including piping and machinery layouts, stability documentation, fire/safety plans, and GA development.
- Provided **R&D and regulatory support**, ensuring compliance with IMO and class requirements across project deliverables.

Key Projects: ATIR Platform (Magallanes Renovables), 1500 RoPax Ferry *Villa del Teror*

Sabbatical and parental leave | Nov 2015 – Oct 2016

During this time, I completed my master's degree in MREs and spent more time with my family.

DNV Maritime Advisory – Potsdam, Germany

Project Engineer / Naval Architect / R&D & CFD Engineer | Jan 2009 – Oct 2015

- Performed **advanced hydrodynamic analyses** (CFD, hull roughness, trim, MetOcean data) to quantify resistance, propulsion efficiency, and fuel consumption.
- Conducted **CFD-based hull optimisation** for ECO Retrofit & ECO Lines projects, achieving **1–10% fuel savings** across diverse vessel types.
- Ensured **IMO/EU** compliance (EEDI, SEEMP, MARPOL Annex VI) by implementing and monitoring energy-efficiency measures.
- Developed **ECO Assistant tool,** applied optimisation tool and contributed to the ECO Insight fleet performance management platform.

Awards: Lloyd's List Awards, Asia 2012 (ECO Assistant)

Newcruise Yacht Projects & Design – Hamburg, Germany

Junior Naval Architect | May 2007 – Jan 2009

- Contributed to **exterior yacht design**, including 3D modelling, general arrangements, and seakeeping assessments.

Key Projects: M/Y Siren (The World Superyacht Awards 2009), M/Y Sapphire, M/Y Roma

Siport21 SL – Madrid, Spain

Junior Naval Architect | Jan 2007 – May 2007

- Conducted **ship dynamics studies**, mooring projects, and production engineering support.

Grupo SAN JOSE SA – Madrid, Spain

Junior Project Engineer | Mar 2004 – Jun 2005

Education

Master's programme in Marine Renewable Energies – Bureau Veritas Business School & Instituto Marítimo Español (IME), 2016 | Grade: 9.1/10

Diploma of Advanced Studies (DEA), PhD Program – Universidad Politécnica de Madrid (UPM), 2007 – 2011 | Grade: 9.0 | Research project: Seakeeping performance assessment of mega-yachts

Mechanical Engineering Study abroad – TU Berlin | Oct 2005 – Jul 2006

Master's Degree in Naval Architecture and Marine Engineering (Dipl.-Ing.) – Universidad Politécnica de Madrid | Oct 1993 – May 2005 | Grade: 6.0
Ship design, shipbuilding, and marine structures | **Final project**: ROV Offshore Support Vessel

International & Spanish Baccalaureate – Instituto de Bachillerato Ramiro de Maeztu | Grade: 8.0

Technical Skills

- **Engineering & Energy Systems:** Marine propulsion systems, alternative fuels, hydrogen technologies (LOHC, LH_2, NH_3), fuel cells (PEMFC, SOFC), thermodynamic systems
- **Simulation & Modelling:** CFD (FINE™/Marine, STARCCM+, ANSYS AQWA), hydrodynamic analysis, heat transfer modelling
- **Programming & Data Analysis:** Python (NumPy, Matplotlib), PostgreSQL, VBA, MATLAB
- **Engineering Software:** EES, NAPA, Maxsurf, Rhino + Grasshopper, AutoCAD, Inventor

- **Data Visualisation:** Tableau, Power BI, MS Excel
- **Project & Technical Documentation:** MS Project, BlueCielo Meridian

Languages

Spanish – Native | English – C1 | German – C1 | French – A1/A2

Certifications

- BI Specialist and Data Analyst with Python certifications, Stackfuel, Mar 2026 – Jul 2026
- Data Analytics (Essentials), Career Foundry, Dec 2025, Mar 2026
- Data visualisation training, data.europa.eu, Oct 2025
- MEC-Supply Chain Guide on Certification & Approvals (Webinar), Lloyd's Register, Sep 2025
- Installation and O&M of Offshore Renewable Energy systems, WavEC, May 2021
- PMP® / PMI Certification Preparation Course | U.N.E.D., 2019 | Grade: 9.2/10
- Lean management workshop, Porsche Consulting, 2012
- Certificate of recreational sailing and motorboat driving license, Marina Mercante 2003
- Mountain guide training – canyoning, climbing and speleology, INEF-UPM, 2001

Interests

Surfing, sailing, climbing, mountaineering | Member: DAV, Sailing Club Beluso, Surfers Against Sewage | Extensive international travel

Landsberg am Lech, 10.04.2026